

class15_pertussis

Thrisha Praveen

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1. Investigating pertussis cases by year

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called cdc and use ggplot to make a plot of cases numbers over time.

```
library(datapasta)
library(ggplot2)
```

```
cdc <- data.frame(
  Year = c(1922L,
    1923L, 1924L, 1925L, 1926L, 1927L, 1928L,
    1929L, 1930L, 1931L, 1932L, 1933L, 1934L, 1935L,
    1936L, 1937L, 1938L, 1939L, 1940L, 1941L,
    1942L, 1943L, 1944L, 1945L, 1946L, 1947L, 1948L,
    1949L, 1950L, 1951L, 1952L, 1953L, 1954L,
    1955L, 1956L, 1957L, 1958L, 1959L, 1960L,
    1961L, 1962L, 1963L, 1964L, 1965L, 1966L, 1967L,
    1968L, 1969L, 1970L, 1971L, 1972L, 1973L,
    1974L, 1975L, 1976L, 1977L, 1978L, 1979L, 1980L,
    1981L, 1982L, 1983L, 1984L, 1985L, 1986L,
    1987L, 1988L, 1989L, 1990L, 1991L, 1992L, 1993L,
    1994L, 1995L, 1996L, 1997L, 1998L, 1999L,
    2000L, 2001L, 2002L, 2003L, 2004L, 2005L,
    2006L, 2007L, 2008L, 2009L, 2010L, 2011L, 2012L,
    2013L, 2014L, 2015L, 2016L, 2017L, 2018L,
    2019L, 2020L, 2021L, 2022L),
  No..Reported.Pertussis.Cases = c(107473,
    164191, 165418, 152003, 202210, 181411,
    161799, 197371, 166914, 172559, 215343, 179135,
    265269, 180518, 147237, 214652, 227319, 103188,
    183866, 222202, 191383, 191890, 109873,
    133792, 109860, 156517, 74715, 69479, 120718,
    68687, 45030, 37129, 60886, 62786, 31732, 28295,
    32148, 40005, 14809, 11468, 17749, 17135,
    13005, 6799, 7717, 9718, 4810, 3285, 4249,
    3036, 3287, 1759, 2402, 1738, 1010, 2177, 2063,
    1623, 1730, 1248, 1895, 2463, 2276, 3589,
    4195, 2823, 3450, 4157, 4570, 2719, 4083, 6586,
    4617, 5137, 7796, 6564, 7405, 7298, 7867,
    7580, 9771, 11647, 25827, 25616, 15632, 10454,
    13278, 16858, 27550, 18719, 48277, 28639,
    32971, 20762, 17972, 18975, 15609, 18617, 6124,
    2116, 3044)
```

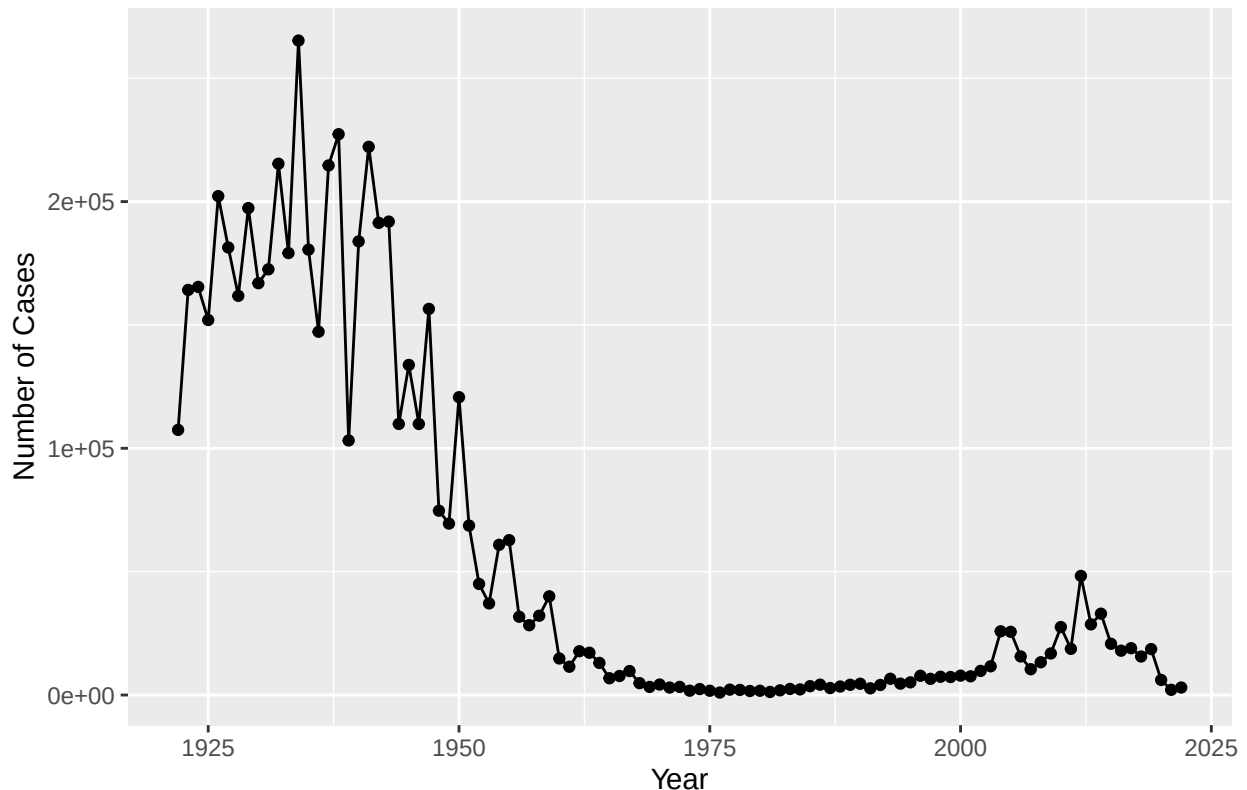
```
)
```

```
head(cdc)
```

```
##   Year No..Reported.Pertussis.Cases
## 1 1922                107473
## 2 1923                164191
## 3 1924                165418
## 4 1925                152003
## 5 1926                202210
## 6 1927                181411
```

```
ggplot(cdc) +
  aes(x = Year, y = No..Reported.Pertussis.Cases) +
  geom_point() +
  geom_line() +
  labs(title = "Pertussis Cases Over Time (1922-2022)", x = "Year", y = "Number of Cases")
```

Pertussis Cases Over Time (1922-2022)

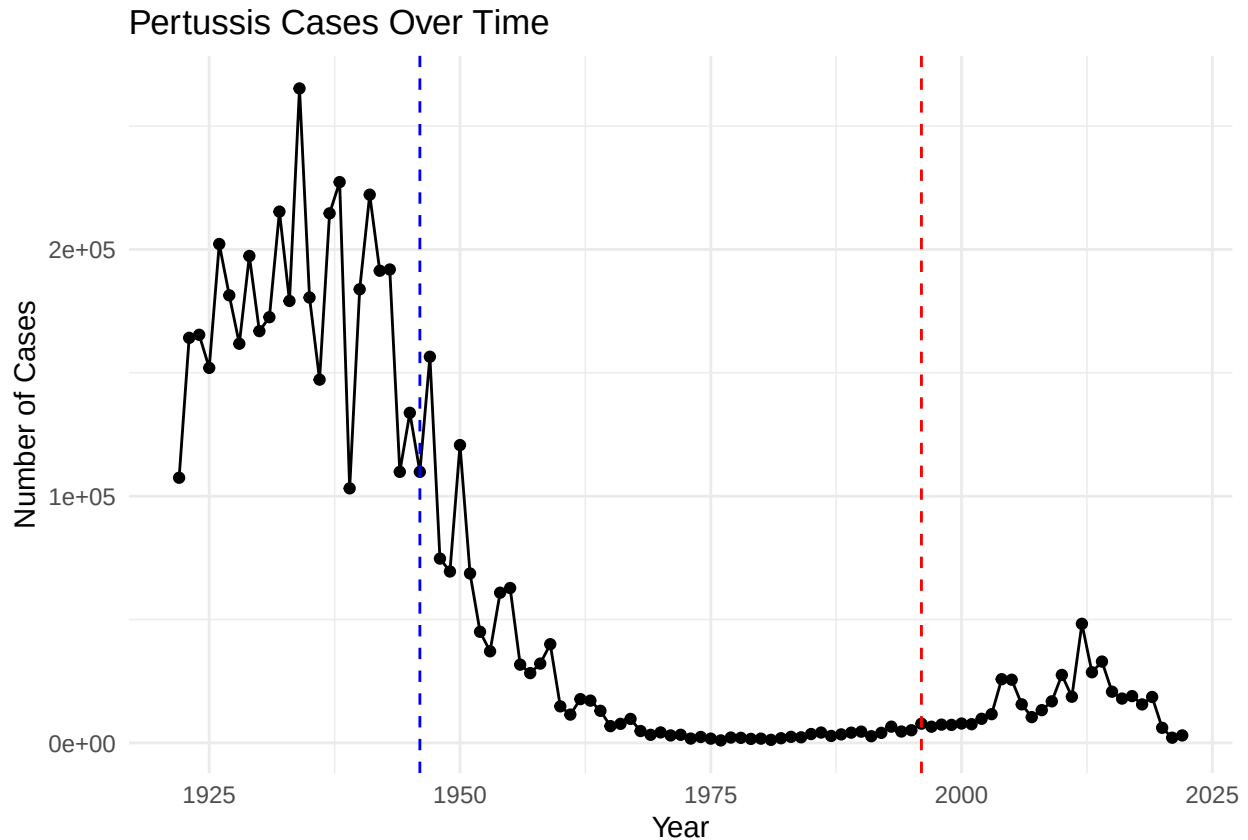


2. A tale of two vaccines (wP & aP)

Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +
  aes(x = Year, y = No..Reported.Pertussis.Cases) +
  geom_point() +
```

```
geom_line() +
geom_vline(xintercept = 1946, linetype = "dashed", color = "blue") +
geom_vline(xintercept = 1996, linetype = "dashed", color = "red") +
labs(title = "Pertussis Cases Over Time",
     x = "Year",
     y = "Number of Cases") +
theme_minimal()
```



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

Pertussis cases are increasing again. A possible explanation is that the bacteria has mutated/evolved enough that it is more resistant to the aP vaccine, making people more susceptible to getting infected.

3. Exploring CMI-PB data

```
# Allows us to read, write and process JSON data
library(jsonlite)
```

```
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
```

```
head(subject, 3)
```

```
##   subject_id infancy_vac biological_sex ethnicity race
## 1          1          wP      Female Not Hispanic or Latino White
## 2          2          wP      Female Not Hispanic or Latino White
```

```
## 3      3      wP      Female      Unknown White
## year_of_birth date_of_boost      dataset
## 1  1986-01-01  2016-09-12 2020_dataset
## 2  1968-01-01  2019-01-28 2020_dataset
## 3  1983-01-01  2016-10-10 2020_dataset
```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
##
## aP wP
## 87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
##
## Female   Male
##   112    60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$biological_sex, subject$race)
```

```
##
##      American Indian/Alaska Native Asian Black or African American
## Female                0    32                2
## Male                  1    12                3
##
##      More Than One Race Native Hawaiian or Other Pacific Islander
## Female                15                1
## Male                   4                1
##
##      Unknown or Not Reported White
## Female                14    48
## Male                   7    32
```

Side note: working with dates

```
library(lubridate)
```

```
##
## Attaching package: 'lubridate'
## The following objects are masked from 'package:base':
##
## date, intersect, setdiff, union
```

```
today()
```

```
## [1] "2025-03-10"
```

```
today() - ymd("2000-01-01")
```

```
## Time difference of 9200 days
time_length( today() - ymd("2000-01-01"), "years")
```

```
## [1] 25.18823
```

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
#subject$age <- as.numeric(today() - ymd(subject$year_of_birth)) / 365
#head(subject)

subject$age <- time_length((today() - ymd(subject$year_of_birth)), "year")
head(subject)
```

```
##  subject_id infancy_vac biological_sex      ethnicity race
## 1           1          wP      Female Not Hispanic or Latino White
## 2           2          wP      Female Not Hispanic or Latino White
## 3           3          wP      Female           Unknown White
## 4           4          wP      Male Not Hispanic or Latino Asian
## 5           5          wP      Male Not Hispanic or Latino Asian
## 6           6          wP      Female Not Hispanic or Latino White
##  year_of_birth date_of_boost      dataset      age
## 1  1986-01-01   2016-09-12 2020_dataset 39.18686
## 2  1968-01-01   2019-01-28 2020_dataset 57.18823
## 3  1983-01-01   2016-10-10 2020_dataset 42.18754
## 4  1988-01-01   2016-08-29 2020_dataset 37.18823
## 5  1991-01-01   2016-08-29 2020_dataset 34.18754
## 6  1988-01-01   2016-10-10 2020_dataset 37.18823
```

```
#int <- today() - ymd(subject$year_of_birth) / 365
#subject$age <- time_length(int, "year")
#head(subject)
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
ap <- subject %>% filter(infancy_vac == "aP")
mean_age <- mean(ap$age, na.rm = TRUE) # Mean
median_age <- median(ap$age, na.rm = TRUE) # Median

print(paste("aP Mean Age:", mean_age))
```

```
## [1] "aP Mean Age: 27.0835752554934"
```

```
print(paste("aP Median Age:", median_age))
```

```
## [1] "aP Median Age: 27.1868583162218"
```

```
wp <- subject %>% filter(infancy_vac == "wP")
mean_age <- mean(wp$age, na.rm = TRUE) # Mean
median_age <- median(wp$age, na.rm = TRUE) # Median

print(paste("wP Mean Age:", mean_age))
```

```
## [1] "wP Mean Age: 35.8342795023554"
```

```
print(paste("wP Median Age:", median_age))
```

```
## [1] "wP Median Age: 34.1875427789185"
```

There is a significant difference between those in the data frame with the wP vaccine and those with the aP vaccine.

Q8. Determine the age of all individuals at time of boost?

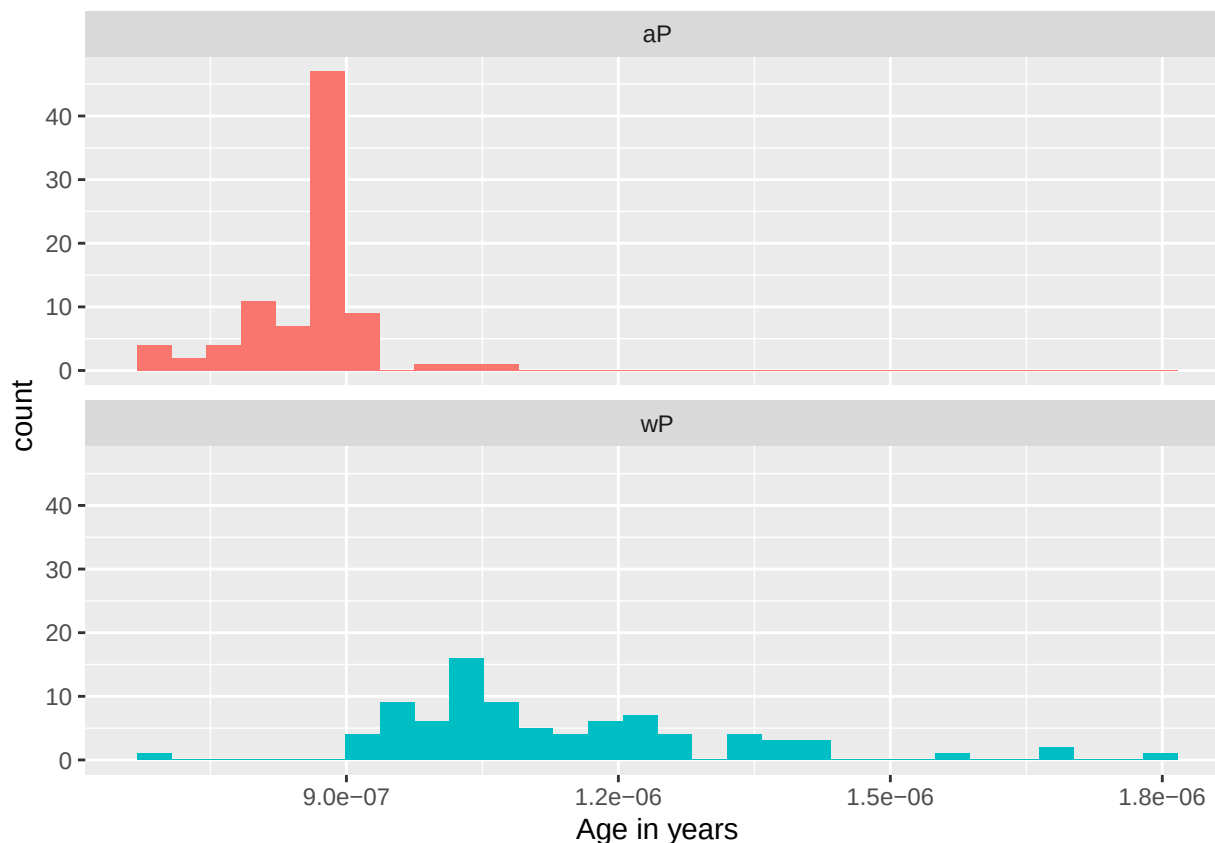
```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
## [1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



^I'm not sure how to fix the x-axis values.

```
x <- t.test(time_length( wp$age, "years" ),
            time_length( ap$age, "years" ))

x$p.value
```

```
## [1] 2.372101e-23
```

Yes, I think these two groups are significantly different.

Joining multiple tables

```
# Complete the API URLs...
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)
titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

Q10. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
meta <- inner_join(specimen, subject)
```

```
## Joining with `by = join_by(subject_id)`
dim(meta)
```

```
## [1] 1503 14
```

```
head(meta)
```

```
## specimen_id subject_id actual_day_relative_to_boost
## 1 1 1 -3
## 2 2 1 1
## 3 3 1 3
## 4 4 1 7
## 5 5 1 11
## 6 6 1 32
## planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
## 1 0 Blood 1 wP Female
## 2 1 Blood 2 wP Female
## 3 3 Blood 3 wP Female
## 4 7 Blood 4 wP Female
## 5 14 Blood 5 wP Female
## 6 30 Blood 6 wP Female
## ethnicity race year_of_birth date_of_boost dataset
## 1 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 2 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 3 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 4 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 5 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 6 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## age
## 1 39.18686
## 2 39.18686
## 3 39.18686
## 4 39.18686
## 5 39.18686
## 6 39.18686
```

Q11. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(titer, meta)

## Joining with `by = join_by(specimen_id)`
dim(abdata)

## [1] 61956 21
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)

##
## IgE IgG IgG1 IgG2 IgG3 IgG4
## 6698 7265 11993 12000 12000 12000
```

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```



```
##
## 2020_dataset 2021_dataset 2022_dataset 2023_dataset
##           31520           8085           7301           15050
```

There was less data in 2021 and 2022, but it has increased for 2023.

4. Examine IgG Ab titer levels

```
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

```
##   specimen_id isotype is_antigen_specific antigen      MFI MFI_normalised
## 1           1    IgG                TRUE      PT  68.56614      3.736992
## 2           1    IgG                TRUE      PRN 332.12718      2.602350
## 3           1    IgG                TRUE      FHA 1887.12263     34.050956
## 4          19    IgG                TRUE      PT   20.11607      1.096366
## 5          19    IgG                TRUE      PRN 976.67419      7.652635
## 6          19    IgG                TRUE      FHA   60.76626      1.096457
##   unit lower_limit_of_detection subject_id actual_day_relative_to_boost
## 1 IU/ML                0.530000           1                    -3
## 2 IU/ML                6.205949           1                    -3
## 3 IU/ML                4.679535           1                    -3
## 4 IU/ML                0.530000           3                    -3
## 5 IU/ML                6.205949           3                    -3
## 6 IU/ML                4.679535           3                    -3
##   planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
## 1                0           Blood           1          wP          Female
## 2                0           Blood           1          wP          Female
## 3                0           Blood           1          wP          Female
## 4                0           Blood           1          wP          Female
## 5                0           Blood           1          wP          Female
## 6                0           Blood           1          wP          Female
##   ethnicity race year_of_birth date_of_boost      dataset
## 1 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 2 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 3 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
## 4           Unknown White 1983-01-01 2016-10-10 2020_dataset
## 5           Unknown White 1983-01-01 2016-10-10 2020_dataset
## 6           Unknown White 1983-01-01 2016-10-10 2020_dataset
##   age
## 1 39.18686
## 2 39.18686
## 3 39.18686
## 4 42.18754
## 5 42.18754
## 6 42.18754
```

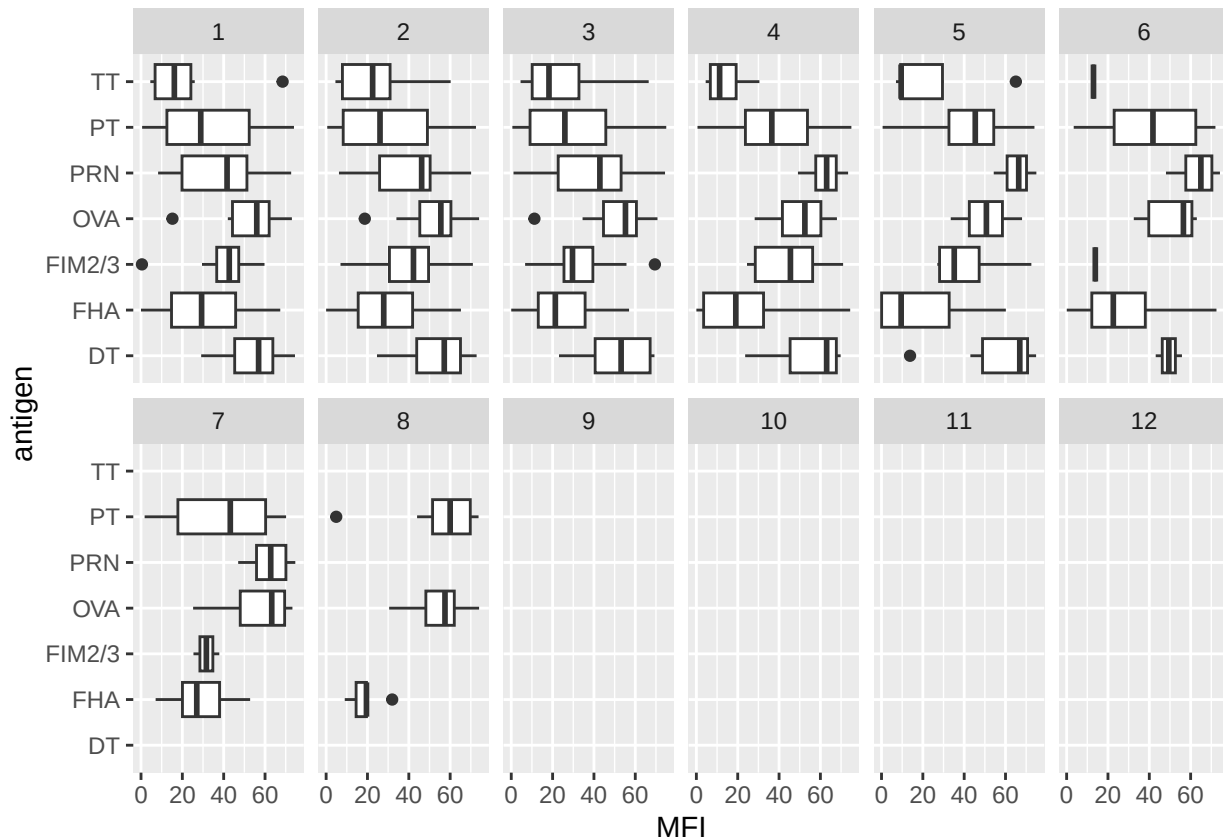
Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
library(ggplot2)

ggplot(igg) +
  aes(MFI, antigen) +
```

```
geom_boxplot() +
xlim(0, 75) +
facet_wrap(vars(visit), nrow = 2)
```

```
## Warning: Removed 6157 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

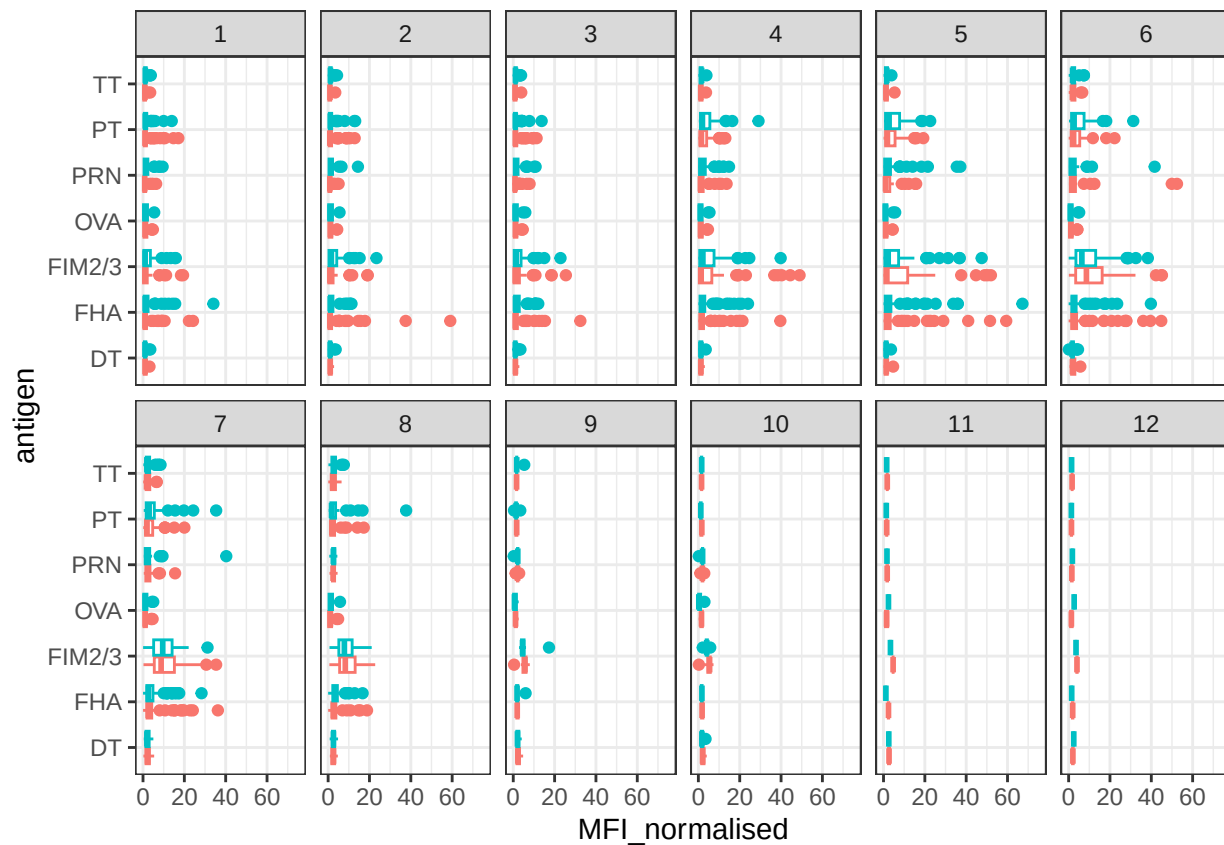


Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

One antigen that has shown a difference in the level of IgG antibody titers recognizing them over time is the PRN antibody (Pertactin autotransporter), which is used to develop acellular vaccines, which are more commonly used in modern times. That's why its recognition has increased over time since more people are getting the aP vaccine. PT (Pertussis toxin) is a key part of the vaccine and thus has also experienced increasing recognition levels.

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

```
## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



```

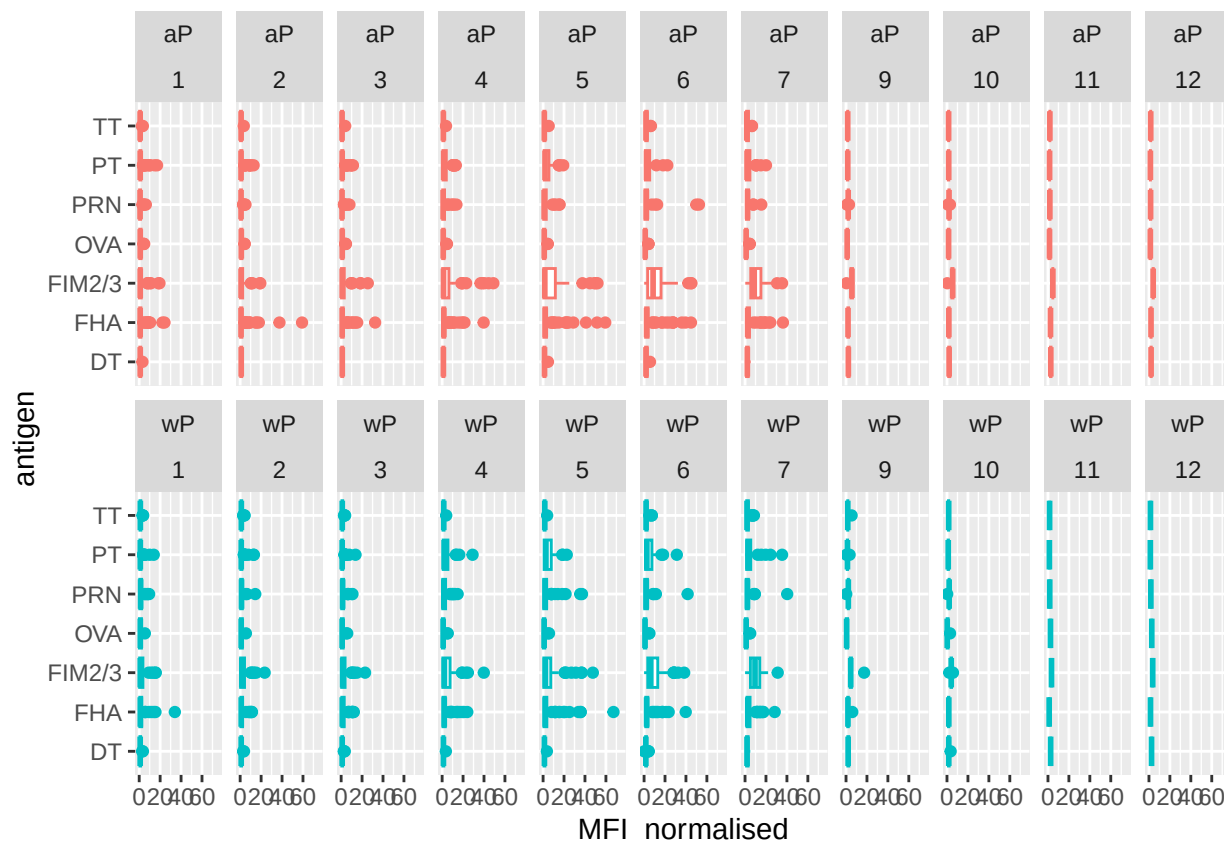
igg %>% filter(visit != 8) %>%
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2)

```

```

## Warning: Removed 5 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

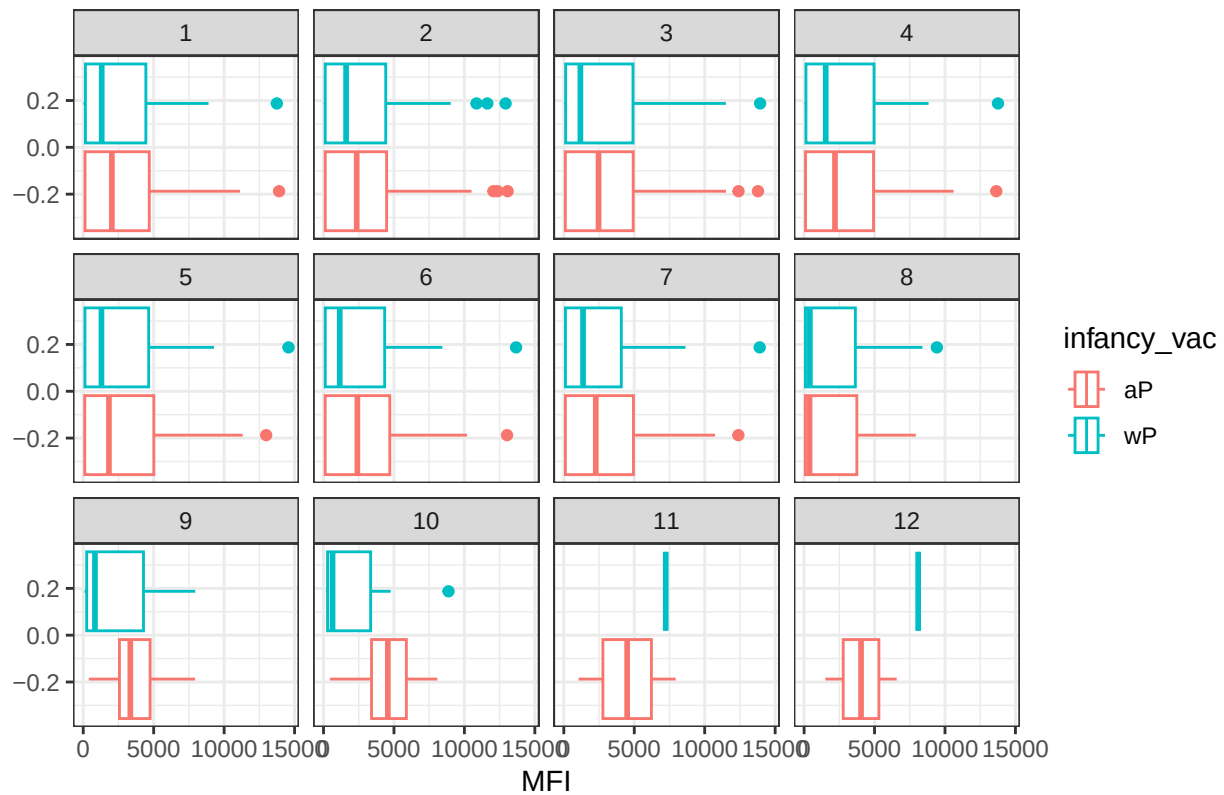
```



Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can choose any you like. Below I picked a “control” antigen (“OVA”, that is not in our vaccines) and a clear antigen of interest (“PT”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).

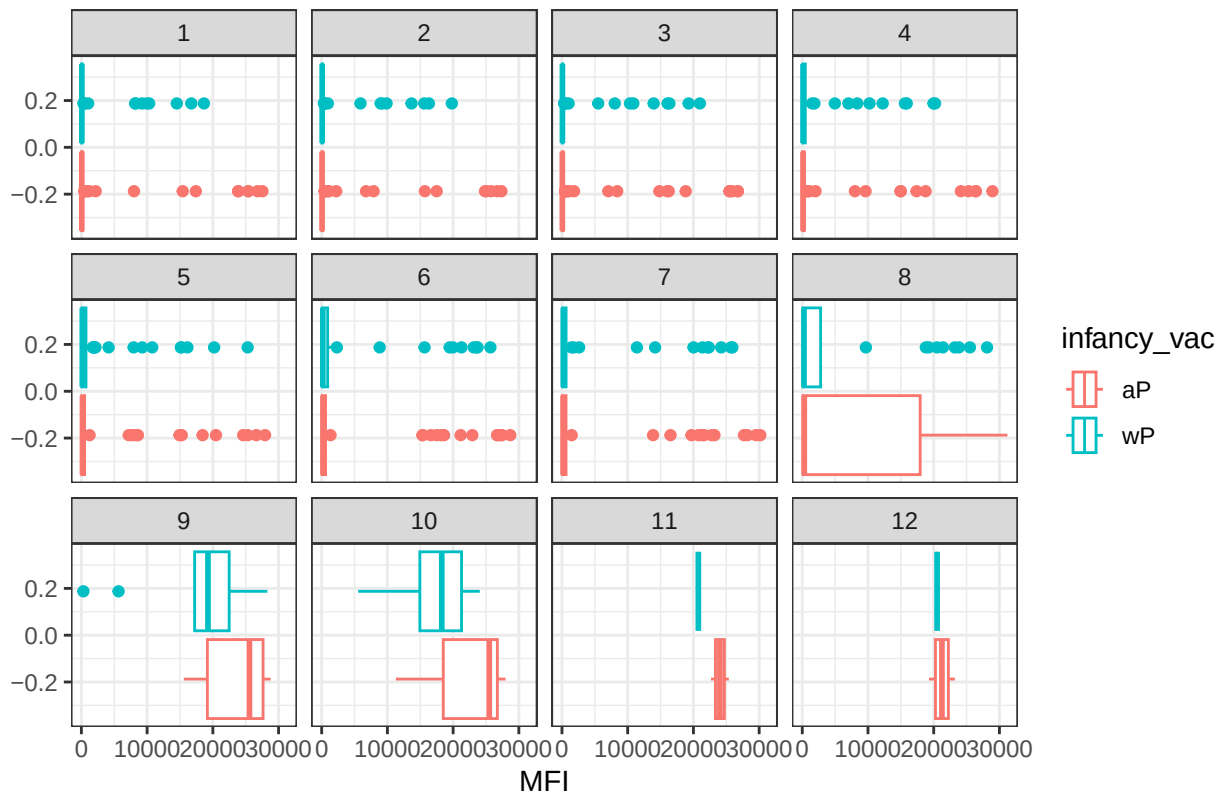
```
filter(igg, antigen=="OVA") %>%
  ggplot() +
  aes(MFI, col=infancy_vac) + # Fill in MFI for the variable to be plotted
  geom_boxplot(show.legend = TRUE) + # If you don't want the legend to show
  facet_wrap(vars(visit)) +
  ggtitle("OVA antigen levels per visit (aP red wP teal)") +
  theme_bw()
```

OVA antigen levels per visit (aP red wP teal)



```
filter(igg, antigen=="PT") %>%
  ggplot() +
  aes(MFI, col=infancy_vac) + # Fill in MFI for the variable to be plotted
  geom_boxplot(show.legend = TRUE) + # If you don't want the legend to show
  facet_wrap(vars(visit)) +
  ggtitle("PT antigen levels per visit (aP red wP teal)") +
  theme_bw()
```

PT antigen levels per visit (aP red wP teal)



Q16. What do you notice about these two antigens time courses and the PT data in particular?

The PT levels increase over time. The OVA levels have only slightly increased over time.

Q17. Do you see any clear difference in aP vs. wP responses?

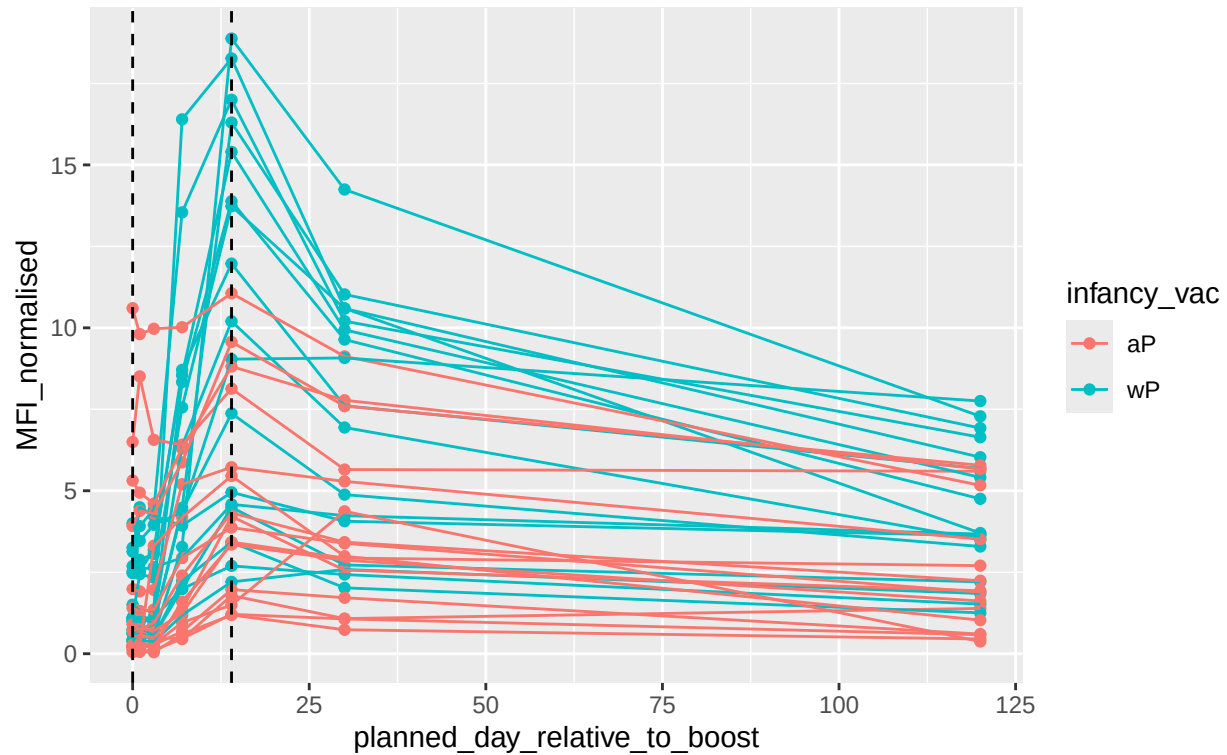
This trend appears similar between wP and aP subjects with PT levels. However, wP patients had a sudden jump in OVA levels in visits 11 and 12.

```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
         y=MFI_normalised,
         col=infancy_vac,
         group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2021 dataset IgG PT",
         subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



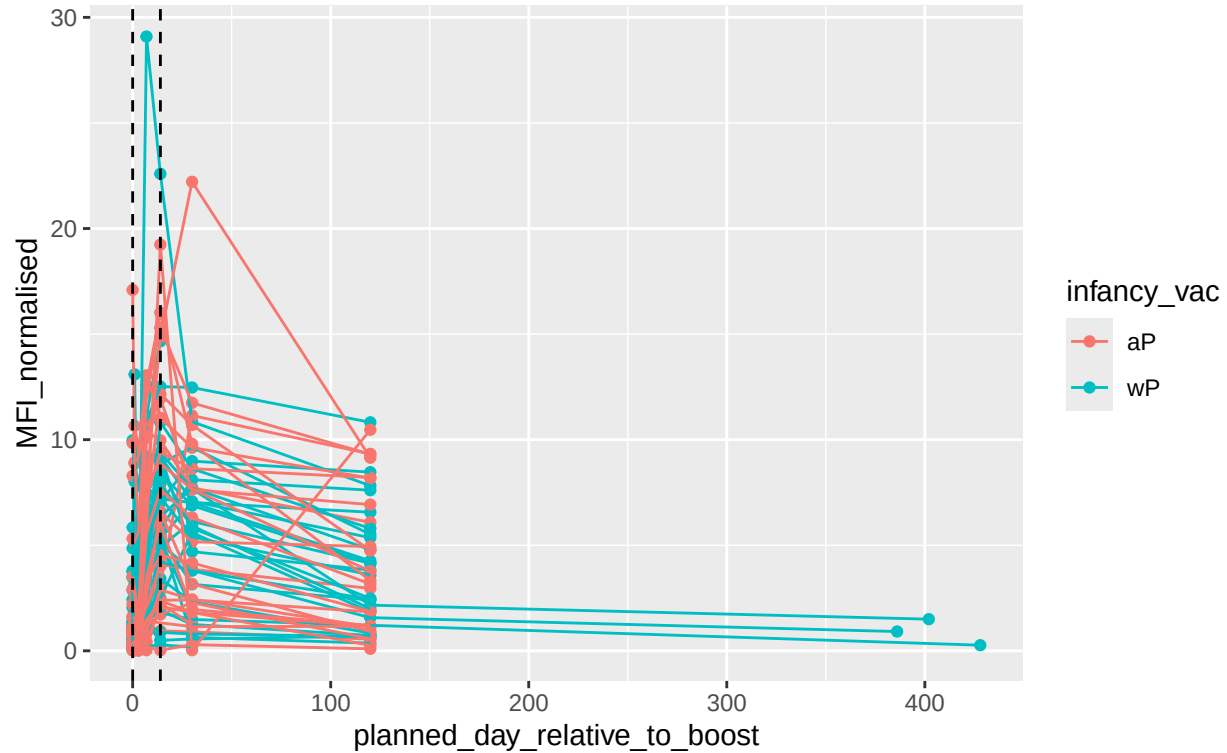
Q18. Does this trend look similar for the 2020 dataset?

```
abdata.20 <- abdata %>% filter(dataset == "2020_dataset")

abdata.20 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
         y=MFI_normalised,
         col=infancy_vac,
         group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2020 dataset IgG PT",
         subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

2020 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



Not exactly, in the 2020 dataset the wP patients seem to have had lower levels of PT antigens than in 2021.

5. Obtaining CMI-PB RNASeq data

```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSOG00000211896.7"
rna <- read_json(url, simplifyVector = TRUE)

meta <- inner_join(specimen, subject)

## Joining with `by = join_by(subject_id)`
ssrna <- inner_join(rna, meta)

## Joining with `by = join_by(specimen_id)`
head(ssrna)

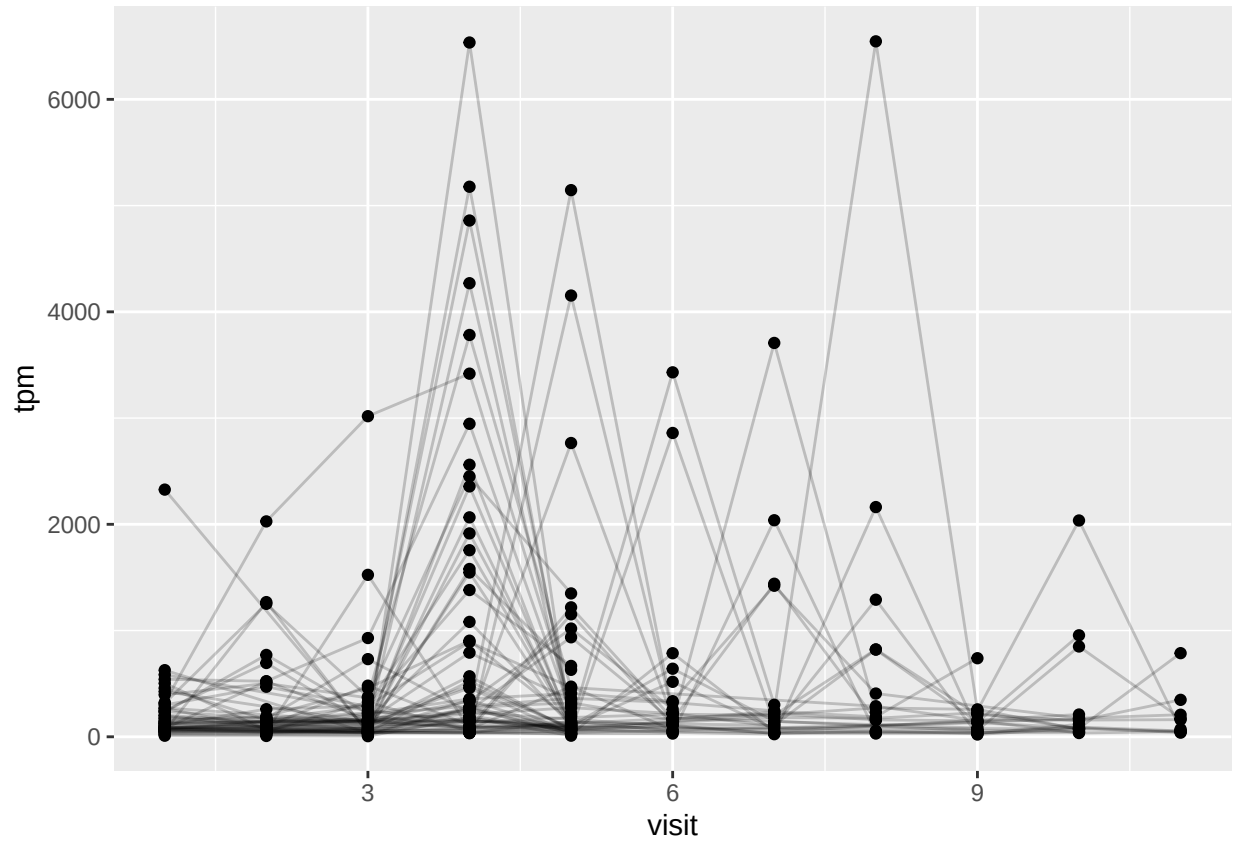
##   versioned_ensembl_gene_id specimen_id raw_count      tpm subject_id
## 1      ENSOG00000211896.7         344    18613  929.640        44
## 2      ENSOG00000211896.7         243     2011  112.584        31
## 3      ENSOG00000211896.7         261     2161  124.759        33
## 4      ENSOG00000211896.7         282     2428  138.292        36
## 5      ENSOG00000211896.7         345    51963 2946.136        44
## 6      ENSOG00000211896.7         244    49652 2356.749        31
##   actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
## 1                           3                           3          Blood
```



```
## 2          3          3          Blood
## 3         92        120          Blood
## 4          1          1          Blood
## 5          7          7          Blood
## 6          7          7          Blood
##   visit infancy_vac biological_sex ethnicity          race
## 1     3          aP      Female   Hispanic or Latino More Than One Race
## 2     3          wP      Female Not Hispanic or Latino          Asian
## 3     7          wP      Male    Hispanic or Latino More Than One Race
## 4     2          aP      Female   Hispanic or Latino          White
## 5     4          aP      Female   Hispanic or Latino More Than One Race
## 6     4          wP      Female Not Hispanic or Latino          Asian
##   year_of_birth date_of_boost   dataset    age
## 1  1998-01-01   2016-11-07 2020_dataset 27.18686
## 2  1989-01-01   2016-09-26 2020_dataset 36.18617
## 3  1990-01-01   2016-10-10 2020_dataset 35.18686
## 4  1997-01-01   2016-10-24 2020_dataset 28.18617
## 5  1998-01-01   2016-11-07 2020_dataset 27.18686
## 6  1989-01-01   2016-09-26 2020_dataset 36.18617
```

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
ggplot(ssrna) +
  aes(visit, tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```



Q20: What do you notice about the expression of this gene (i.e. when is it at it's maximum level)? > It's highest during visits 4-8.

Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

As time passes, the antibody levels increase and then slowly decrease. When they decrease, the antigen levels increase, which matches what's expected.